Scalar inequalities for subdivision counts

Supplementary material

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Supplement to *Mean order and counts of dominating sets in graphs of small degree*

1 Scalar inequality for subdivision counts

This note proves the scalar inequality used in the once-subdivided-core count proposition of the accompanying article. All logarithms are natural. Fix real $d \geq 3$ and $0 < q < 1$, and write

$$x = dq^2, \quad r = 1 + x, \quad p = q/r, \quad L = -\log(1-q), \quad T = -\log(1 - 4q^3 + 3q^4).$$

The quantity to be bounded is

$$\Phi_d(q) = \frac{d+2}{8}T - \mathrm{KL}(p\|q) + p\log(1-q^d) - \frac{dp^2L}{2},$$

where KL denotes binary relative entropy. We also write $h(p) = -p\log p - (1-p)\log(1-p)$ for the binary entropy. The argument below proves $\Phi_d(q) > 0$ on the whole stated domain. The real-d formulation is convenient for the certificates; graph degrees are integers.

1.1 The interval $0 < q \leq 1/4$

The integral representation

$$\mathrm{KL}(p\|q) = \int_p^q \frac{t-p}{t(1-t)}\,dt$$

and $\log(1+x) \geq 2x/(2+x)$ give

$$\mathrm{KL}(p\|q) \leq \frac{qx^2}{r(x+2)} + \frac{q^2x^2}{2r^2(1-q)}.$$

To see this, split $1/[t(1-t)] = 1/t + 1/(1-t)$. The first integral is $q[x - \log(1+x)]/r$; in the second, bound $1/(1-t)$ by $1/(1-q)$. Moreover,

$$T \geq 4q^3 - 3q^4, \qquad L \leq q + \frac{q^2}{2(1-q)}, \qquad p\log(1-q^d) \geq -\frac{q^4}{r(1-q^3)}.$$

Substitution gives

$$\frac{r^2\Phi_d(q)}{q} \geq \frac{x^2(x^2+2x+2)}{2(x+2)} - \frac{3xqr^2}{8} + q^2r^2\left(1 - \frac{3q}{4}\right) - \frac{qx^2}{2(1-q)} - \frac{xq}{4(1-q)} - \frac{q^3r}{1-q^3}.$$

Use $x^2(x^2+2x+2)/[2(x+2)] \geq x^2/2 + x^3/4$, $1/(1-q) \leq 4/3$, $1/(1-q^3) \leq 64/63$, and $1 - 3q/4 \geq 13/16$. The resulting lower bound is

$$\left(\frac{1}{4} - \frac{3q}{8}\right) x^3 + a(q)x^2 - b(q)qx + c(q)q^2,$$

where

$$a(q) = \frac{1}{2} - \frac{17q}{12} + \frac{13q^2}{16},$$
$$b(q) = \frac{17}{24} - \frac{13q}{8} + \frac{64q^2}{63}, \qquad c(q) = \frac{13}{16} - \frac{64q}{63}.$$

All three are positive here, and the cubic coefficient is at least $5/32$. For $q \leq 1/8$, the bounds $a \geq 31/96$, $c \geq 691/1008$ and $b \leq 17/24$ give $4ac - b^2 \geq 9283/24192 > 0$. For $1/8 \leq q \leq 1/4$, use $a \geq 7/48$, $c \geq 563/1008$ and $b \leq 2293/4032$ to obtain $4ac - b^2 \geq 38855/16257024 > 0$. Thus the quadratic is positive, proving the claim on this interval.

1.2 The interval $1/4 \leq q < 1/2$

Put $w = q^3(4-3q)$ and $y = x/(x+2)$. Keeping additional terms gives

$$T \geq w + w^2/2, \qquad L \leq q + q^2/2 + \frac{q^3}{3(1-q)},$$
$$\log(1-q^d) \geq -q^3 - \frac{q^6}{2(1-q^3)},$$
$$\mathrm{KL}(p\|q) \leq \frac{q}{r}\left[x - 2\left(y + \frac{y^3}{3}\right)\right] + \frac{q^2x^2}{6r^2}\left[\frac{r}{r-q} + \frac{2}{1-q}\right].$$

The logarithmic estimates follow from the power series with geometric tail bounds. In the entropy estimate, use the first two terms of $\log r = 2\,\mathrm{artanh}\, y$ for the $1/t$ integral. For the $1/(1-t)$ integral, convexity puts $1/(1-t)$ below its endpoint secant on $[p,q]$; integrating that secant against $t-p$ gives the displayed term.

Let $B(q,x)$ be the resulting rational lower bound, with $d = x/q^2$:

$$B = \frac{x/q^2+2}{8}(w + w^2/2) - \frac{q}{r}\left[x - 2\left(y + \frac{y^3}{3}\right)\right]$$
$$- \frac{q^2x^2}{6r^2}\left[\frac{r}{r-q} + \frac{2}{1-q}\right] - \frac{q}{r}\left[q^3 + \frac{q^6}{2(1-q^3)}\right]$$
$$- \frac{x}{2r^2}\left[q + \frac{q^2}{2} + \frac{q^3}{3(1-q)}\right].$$

Every multiplication preserves the indicated direction, so $\Phi_d(q) \geq B$. The positive denominator

$$D_1 = 48r^2(x+2)^3(r-q)(1-q)(1-q^3)$$

clears B/q: $N_1(q,x) = D_1 B(q,x)/q$ is a polynomial with degrees $(12,7)$ in (q,x). Substitute

$$x = 3q^2 + s, \qquad q = \frac{1+2u}{4(1+u)}, \qquad s,u \geq 0.$$

Then $M_1(u,s) = [4(1+u)]^{23} N_1(q, 3q^2+s)$ has *192 nonzero coefficients, all positive*, and constant coefficient 794008104013215. The coefficient list is supplied with an exact reconstruction program. For each $d \geq 3$ and $1/4 \leq q < 1/2$, the inverse parameters are $s = (d-3)q^2 \geq 0$ and $u = (4q-1)/(2-4q) \geq 0$. Thus $M_1 > 0$, whence $N_1 > 0$, $B > 0$ and $\Phi_d(q) > 0$. The next interval includes $q = 1/2$, so no limit at $u = \infty$ is required.

1.3 The interval $1/2 \le q < 1$

The separate logarithmic estimates above are not useful near $q = 1$. Instead, cancel the divergent terms first. Set

$$V = 1 + 2q + 3q^2, \qquad S_d = \frac{1 - q^d}{1 - q}, \qquad c_0 = \frac{d-2}{4} - \frac{dp^2}{2}.$$

Factoring $1 - 4q^3 + 3q^4 = (1 - q)^2 V$ gives the exact identity

$$\Phi_d(q) = c_0 L + h(p) + p \log(qS_d) - \frac{d+2}{8} \log V. \tag{1}$$

Since $dp^2 = x/(1 + x)^2 \le 1/4$, we have $c_0 \ge 1/8$. Also $S_d \ge 1 + q + q^2$ for real $d \ge 3$. Put

$$U = (r - q)(1 + q + q^2), \qquad \eta = \frac{U - 1}{U + 1}.$$

Here $U > 1$, since $r - q \ge 5/4$ and $1 + q + q^2 \ge 7/4$. The identity

$$h(p) + p \log(qS_d) = -\log(1 - p) + p \log[(r - q)S_d]$$

therefore yields

$$h(p) + p \log(qS_d) \ge p + p^2/2 + 2p(\eta + \eta^3/3).$$

For $a = q/(2 - q)$ and $b = (V - 1)/(V + 1)$, both in $(0, 1)$, define

$$L_- = 2 \sum_{j=0}^{3} \frac{a^{2j+1}}{2j + 1},$$

$$V_+ = 2 \left\{ \sum_{j=0}^{3} \frac{b^{2j+1}}{2j + 1} + \frac{b^9}{9(1 - b^2)} \right\}.$$

The atanh series gives $L \ge L_-$ and $\log V \le V_+$. Because $c_0 > 0$, (1) implies

$$\Phi_d(q) \ge C(q, x) := c_0 L_- + p + p^2/2 + 2p(\eta + \eta^3/3) - \frac{d+2}{8} V_+.$$

For explicit polynomial clearing, write $a_0 = 2 - q$, $W = V + 1$ and $U_\pm = U \pm 1$, and set

$$L_n = \sum_{j=0}^{3} \frac{210}{2j + 1} q^{2j+1} a_0^{6-2j},$$

$$V_n = \sum_{j=0}^{3} \frac{1260}{2j + 1} V (V - 1)^{2j+1} W^{6-2j} + 35(V - 1)^9,$$

$$C_n = (x - 2q^2) r^2 - 2xq^2,$$

$$E_n = 2qr\{3U_+^3 + 6U_- U_+^2 + 2U_-^3\} + 3q^2 U_+^3.$$

Then $L_- = L_n/(105a_0^7)$, $V_+ = V_n/(630VW^7)$, $c_0 = C_n/(4q^2r^2)$, and the entropy lower bound is $E_n/(6r^2U_+^3)$. The positive factor $D_2 = 5040q^2r^2a_0^7VW^7U_+^3$ consequently gives

$$P(q, x) := D_2 C(q, x) = 12 C_n L_n V W^7 U_+^3 + 840 q^2 a_0^7 V W^7 E_n - (x + 2q^2) V_n r^2 a_0^7 U_+^3.$$

After $x = 3q^2 + s$, substitute $q = (1 + 2u)/[2(1 + u)]$ and define

$$M_2(u, s) = [2(1 + u)]^{43} P(q, 3q^2 + s).$$

This polynomial has degrees $(43, 6)$ in (u, s) and *308 nonzero coefficients, all positive.* Its constant is 2916867824861087696175. For every $d \ge 3$ and $1/2 \le q < 1$, $s = (d - 3)q^2 \ge 0$ and $u = (2q - 1)/(2 - 2q) \ge 0$ are finite, so $M_2 > 0$, $P > 0$, $C > 0$ and $\Phi_d(q) > 0$. Together the three intervals prove the scalar inequality.

1.4 Verification

The complete coefficient lists for M_1 and M_2 accompany this note. Separate Python programs reconstruct both polynomials using exact integer arithmetic and check every coefficient. The supplementary verification guide identifies the files and gives the commands for these calculations. The denominator signs and domain substitutions above complete the reduction from coefficient positivity to the stated scalar inequality.

The result concerns counts. For cores of degree three or four and $t \geq 1$, strictness also follows from the main article's equality classification. The present argument covers arbitrary core degrees at least three and all positive activities on the once-subdivision class.